T-flex

FLEXIBLE PIPE SYSTEM FOR WATER SUPPLY AND SEVERAL APPLICATIONS



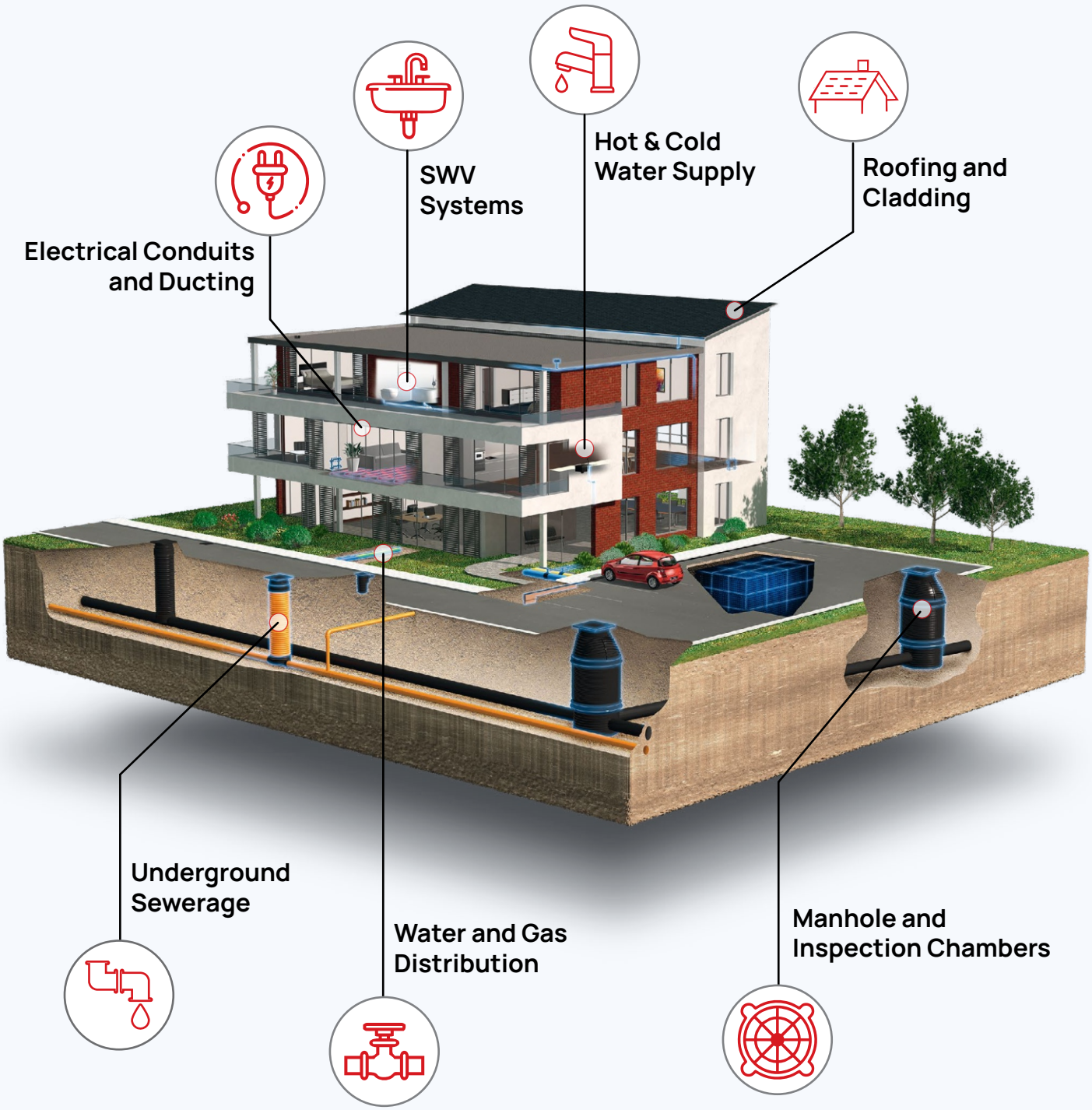


HDPE

DADEX

DADEX

Leader in Roofing and Thermoplastic Piping Systems



Discover our broad portfolio at

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About Dadex

Dadex was established as a public limited company in 1959. We started with the manufacturing of FC roofing systems at our facilities in Hyderabad, under the dynamic leadership of our Founder Chairman, Mr. Kassim Dada. Being a forward-looking organization, his son, Mr. Sikander Dada kept up to date with the market, and was instrumental in the diversification with plastic piping systems. Presently we produce various thermoplastic systems such as UPVC, PP-R, PP-RCT, PP, PE.

Headquartered in Karachi, our two manufacturing facilities based in Hyderabad and Sundar Industrial Estate (near Lahore) are equipped with state of the art machinery. Our joint efforts have led Dadex to the ISO 9001, ISO 14001, ISO 45001 and PSQCA certifications.

We aim to set standards in every field we enter. Customer focus, quality and innovation are reflected in every aspect of our business. Offering customized solutions to a cross section of markets for over 66 years, Dadex today is a hallmark of trust and reliability.

We believe that our strength lies in delighting our customers. That is why providing quality products backed by round the clock technical support services are the Dadex promise to all our valued customers.

Extensive Product Range

Water Supply - Infra-Structure

- Aquadex (UPVC Pressure Pipe System)
- T-Flex (PE 100 Pipes & Fittings)

Water Supply - Housing

- Polydex - Premium (PP-RCT Pipe System)
- Polydex (PP-R Pipe System)
- Thermoline (PP-R Pipe System)
- Aquadex (UPVC Pressure Pipe System)

Soil, Waste and Vent (SWV)

- Nikasi - Solvent (UPVC Pipe System)
- Nikasi - Rubber Ring (UPVC Pipe System)
- Polyduct (Low Noise System)

Underground Sewerage and Drainage

- Flow Line (UPVC Underground Sewerage System)
- Inspection Chambers and Manholes

Cable Ducting

- PE Cable Duct Pipes
- Electroduct (UPVC Conduits as per IEC - EN Standard)
- Conduits (UPVC Conduits as per Dadex Standard)

Gas Supply

- T-Flex for gas (PE 80 and PE 100)

Introduction

T-flex pipe systems are designed for many pressure and non pressure applications such as water supply and distribution, compressed air, chemicals, hazardous wastes, utility, slurries, marine, mining, and agriculture. T-flex pipe systems offered by Dadex are manufactured from polyethylene (PE) compound.
That is why T-flex provides professionals an option to choose the best system for the most demanding projects.

T-flex pipes have excellent abrasion resistance, superb impact resistance, and extraordinary toughness. These pipes are resistant to a broad range of corrosive chemicals; they do not support biological growth, and resist the adherence of scale and deposits.

Material

T-Flex pipes are made from PE 100 black compound sourced from the finest manufactures.

Product Range

HDPE T-Flex pipes are manufactured in nominal outside diameter of:
20, 25, 32, 40, 50, 63, 75, 90, 110, 125,160, 180, 200, 250, 280,
315, 355, 400, 450, 500, 560 and 630mm.



Standard Lengths

Coils up to 50m and 100m lengths are available for sizes up to 90mm.
Pipes are available in straight lengths of 6m and 12m up to 630mm.

Identification Stripes and Colour

T-flex pipes are commonly manufactured in black colour with blue / yellow coloured stripes or black colour only without stripes as per standard application. Standard Dadex marking identify application (water, gas, cable duct, etc.)

Physical Dimension of PE Pipes as per DIN 8074/8075 / ISO 4427-2:2007(E) / PS-3580:1997

Nominal Size O.D (mm)	PN 6 / SDR 26		PN 8 / SDR 21		PN 10 / SDR 17		PN 12.5 / SDR 13.6		PN 16 / SDR 11	
	Wall thickness		Wall thickness		Wall thickness		Wall thickness		Wall thickness	
	Min (mm)	Max (mm)	Min (mm)	Max (mm)	Min (mm)	Max (mm)	Min (mm)	Max (mm)	Min (mm)	Max (mm)
20	-	-	-	-	-	-	-	-	2.0	2.3
25	-	-	-	-	-	-	2.0	2.3	2.3	2.7
32	-	-	-	-	2.0	2.3	2.4	2.8	3.0	3.4
40	-	-	2.0	2.3	2.4	2.8	3.0	3.5	3.7	4.2
50	2.0	2.3	2.4	2.8	3.0	3.4	3.7	4.2	4.6	5.2
63	2.5	2.9	3.0	3.4	3.8	4.3	4.7	5.3	5.8	6.5
75	2.9	3.3	3.6	4.1	4.5	5.1	5.6	6.3	6.8	7.6
90	3.5	4.0	4.3	4.9	5.4	6.1	6.7	7.5	8.2	9.2
110	4.2	4.8	5.3	6.0	6.6	7.4	8.1	9.1	10.0	11.1
125	4.8	5.4	6.0	6.7	7.4	8.3	9.2	10.3	11.4	12.7
160	6.2	7.0	7.7	8.6	9.5	10.6	11.8	13.1	14.6	16.2
180	6.9	7.7	8.6	9.6	10.7	11.9	13.3	14.8	16.4	18.2
200	7.7	8.6	9.6	10.7	11.9	13.2	14.7	16.3	18.2	20.2
250	9.6	10.7	11.9	13.2	14.8	16.4	18.4	20.4	22.7	25.1
280	10.7	11.9	13.4	14.9	16.6	18.4	20.6	22.8	25.4	28.1
315	12.1	13.5	15.0	16.6	18.7	20.7	23.2	25.7	28.6	31.6
355	13.6	15.1	16.9	18.7	21.1	23.4	26.1	28.9	32.2	35.6
400	15.3	17.0	19.1	21.2	23.7	26.2	29.4	32.5	36.3	40.1
450	17.2	19.1	21.5	23.8	26.7	29.5	33.1	36.6	40.9	45.1
500	19.1	21.2	23.9	26.4	29.7	32.8	36.8	40.6	45.4	50.1
560	21.4	23.7	26.7	29.5	33.2	36.7	41.2	45.5	50.8	56.0
630	24.1	26.7	30.0	33.1	37.4	41.3	46.3	51.1	57.2	63.1

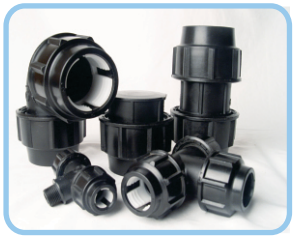
Note: 1 bar = 10.2 m head of water = 0.1 MPa

Technical Specifications

Pressure Rating of HDPE T-flex Pipes:
Max. working pressure of HDPE T-flex pipes range from 6, 8, 10, 12.5 and PN 16 bar. The nominal pressure (PN) corresponds to the maximum allowable working pressure in bar for pipe at 20°C, including pressure surge. Adequate Surge allowance must be taken into account by pipe line designer.

Max. Working Pressure of PP and PE Fittings:
10 bar (Compression Type); 8-16 bars (Butt Fusion Type)

Cold Bending Radii (CBR):
CBR in meters at 20°C = 22 x Outside Diameter of pipe.





Standards

HDPE T-flex pipes for potable water application are manufactured according to the international standard ISO 4427-2(E):2007, PE pipes for water supply specification DIN 8074/8075 and PS – 3580:1997.
Fittings are compatible with ISO (or metric) dimension pipe and conform to the following specifications;
PP Compression fittings: ISO 3458, 3459, 3501, 3503, & BS:5114; PE Butt Fusion fittings: DIN 16963.

Features & Benefits

Flexibility and Toughness: T-flex is flexible, allowing it to follow rolling terrain contours and reducing the need for fittings. It retains working flexibility even in harsh climates. Water within the pipe may freeze without damaging the pipe. Unstable soils and seasonal freeze/thaw conditions have little effect on this flexible, plastic pipe system. It also exhibits high stress/strain sustainability arising due to mechanical & thermal changes.

Non-contaminating (food grade material): The purity of the fluids being conveyed is safeguarded by the absence of easily extractable substances. The raw material used for manufacturing T-flex piping system for potable water has been evaluated and certified by authorities. Potable water products meet the requirements of standards such as ISO/DIN/EN.

Sealed Joints: T-flex pipes can be joined into long, continuous lengths by compression, butt fusion or electro fusion fittings. These jointing techniques provide leak-free joints that are as strong and chemically resistant as the pipe itself.

Light Weight & Easy to Install: T-flex is much lighter than ductile iron, steel and reinforced concrete. Thus it does not require heavy handling and laying equipment otherwise commonly used for conventional pipe systems this makes installation of pipes easier and quicker.

UV Resistant: T-flex contains carbon black as UV stabilizer which is one of the most efficient and widely used UV absorbers. Carbon black converts harmful ultraviolet radiation into harmless infrared radiation or thermal energy, which dissipates through the polymer matrix neutralizing the harmful effect of UV. Thus, T-flex is UV resistant and safe to use in exposed conditions.

Fields of Application

- Acid / Caustic Lines
- Chilled Water Supply
- Crude Oil
- Dredging
- Fertilizer
- Hazardous Waste
- Sea Water Effluents
- Slurries
- Utility Piping
- Compressed Air (buried)
- Water Supply and Distribution
- Cable Conduits
- Cooling Water
- Drain Lines / Industrial Effluents
- Drip Irrigation / Sprinkle Irrigation
- Fly Ash
- Process Piping
- Sludge Piping
- Underground Services
- Gas (Manufactured as per ISO 4437)
- Brine
- De-Watering Pipes
- Drilling Mud
- Coal Slurry
- Fire Water Mains
- Out Fall Pipelines
- Sewage Treatment
- Storage Tank Piping
- Industrial Use

Typical Physical Properties of Polyethylene (Black) Material PE-100

Physical Properties*		Typical Value *	Unit	Test Method
Density	(Compound)	959	kg/m3	ISO 1183/ISO 1872-2B
Melt Flow Rate	(190°C/2.16 kg)	‹0.1	g/10 min	ISO 1133
Melt Flow Rate	(190°C/5.0 kg)	0.25	g/10 min	ISO 1133
Tensile Stress at Yield	(50 mm/min)	25	Mpa	ISO 527-2
Elongation at Break		›600	%	ISO 527-2
Charpy Impact, notched	(0°C)	16	kJ/m2	ISO 179/1eA
Hardness, Shore D		60	-	ISO 868
Carbon Black content		≥2	%	ASTM D 1603
Brittleness Temperature		‹-70	°C	ASTM D 746
Thermal Stability	(210°C)	›15	min	EN 728

*** The data should not be used for specification work.**

Certification

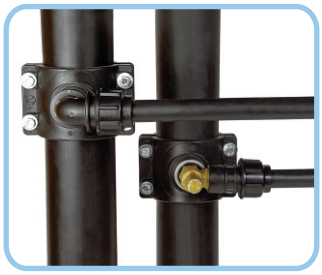
Company meets the requirements according to ISO 9001, ISO 14001 and ISO 45001.

Handling and Storage

General Principles

Polyethylene is a tough resilient material which is relatively light and easy to handle although it is prone to damage through scoring by sharp objects. Therefore careful handling is always required and the dragging of straight pipe and coils should be avoided.

- The maximum allowable depth of scoring of the external surface of the pipe is 10% of the wall thickness. Pipes and fittings showing obvious defects or excessive scoring should be withdrawn, clearly identified as unsuitable and, where appropriate, returned to the source of supply.
- The general properties of polyethylene are unaffected by low ambient temperatures but having very smooth surfaces, the pipes and fittings become slippery in wet or frosty weather. Particular attention should be given to effective securing and storage under such conditions.



The rules on handling and storage

Never

- Drag or roll individual pipes or bundles.
- Throw/drop pipe/fittings from delivery vehicles.
- Use metal slings, hooks or chains when handling.
- Stack more than 2.5 meters or two bundles high.
- Expose pipe/fittings to prolonged sunlight. (Protect with opaque sheeting or tarpaulin).
- Place pipes or fittings in contact with lubricating or hydraulic oils, gasoline, solvents or other aggressive materials.

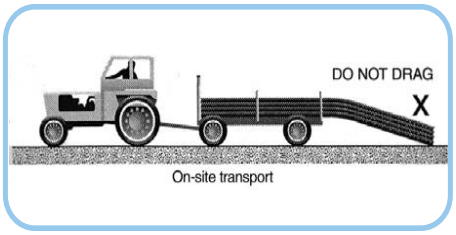
Always

- Store pipes on flat, firm ground, able to withstand weight of pipe/fittings and lifting apparatus.
- Keep pipe/fittings well away from sharp objects, such as flints.
- Use wide non-metallic slings (e.g. nylon or polypropylene).
- Exercise care when handling pipes in wet or frosty conditions, since they may become slippery.
- Keep protective packaging (battens, shrinkwrap, pallets, strapping, etc.) intact until pipes/fittings are required for use.
- Keep pipes/fittings away from intense heat, except when jointing.
- Allow for some bending deflection when pipes are loaded and unloaded. Lifting points should be evenly spaced.

Coiled Pipes

Pipe sized >63mm should be moved and uncoiled using an approved dispensing trailer. Before unstrapping pipe from the coil or drum, both pipe ends must be firmly mechanically restrained. The band securing the outer end of the pipe should be removed first and the movement of the free end carefully controlled. This removal should be followed with those securing successive layers. No more bands should be removed than necessary to release the length of pipe immediately required. When removed from the coil or drum, the pipe will be oval and curved. The extend of the ovality and curvature will depend upon temperature, SDR rating, pipe diameter, coil diameter and material type. Although both ovality and curvature will reduce naturally with time, special re-rounding tools are available to facilitate handling and jointing.

Loading and Off-loading



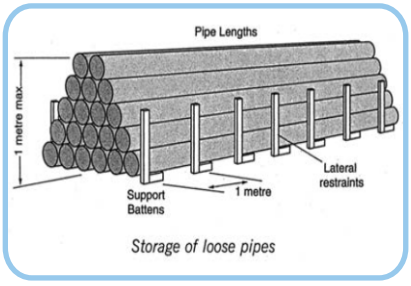
On-site transport

Lengths and bundles

A flat-bed vehicle, free from sharp objects and projections should be used for transportation of pipe systems. When lifting pipe bundles by crane, wideband slings must be used; do not use chains, hooks or wire slings. Do not drag pipe across truck deck or the ground. For lengths greater than six meters, load spreading beams should be inserted at equal distances apart.

Care should be taken to avoid positioning pipes and fittings near or adjacent to exhaust systems or other head sources and to avoid possible contamination from materials such as diesel oil.

Storage



Storage of loose pipes

Care should be taken to avoid positioning pipes and fittings near or adjacent to exhaust systems or other head sources and to avoid possible contamination from materials such as die. All pipes and fittings should be inspected prior to storage and any damaged items isolated and removed from stock. The supplier should be notified immediately of any defective product. As PE pipes are date stamped at manufacturing time, stocks should be arranged so that the earliest date production is used first in installation.

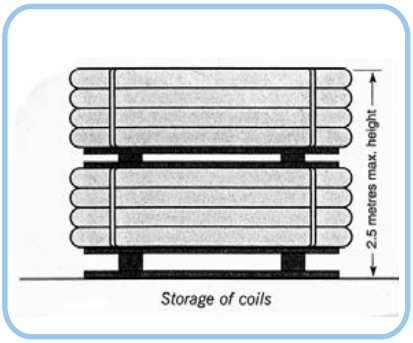
The same procedure should be followed with fittings, where the packaging indicates a date of manufacture. Where pipes are stored on site, the ground should be flat, and free from all rocks. Pipes must not be stored near high temperature sources and must be kept away from combustible materials and potential contaminants. Under no circumstance should coils be stored higher than 2.5m.

Fittings

Electrofusion fittings should be stored under cover in dry conditions, preferably on racking. They should be kept in their boxes and packaging until ready for use. Butt fusion and spigotted fittings may be stored outdoors, as long as they are protected against damage and prolonged direct sunlight. Electrofusion fittings should be retained in their plastic bags until used.

Lengths

Pipe lengths stored individually should be stacked in a pyramid not more than one meter high, with the bottom layer fully restrained by wedges. Where possible, the bottom layer of pipes should be laid on timber battens at one-metre centres. On site, pipes may be laid out individually in strings. (Where appropriate, protective barriers should be placed with adequate warning signs and lamps.)



Storage of coils

Design Consideration

1. Where PE Pressure Pipes are selected, the designer must consider:

- The use of straight or coiled pipes
- Rehabilitation
- The jointing method
- Trench Design

2. PE Pressure pipes are available either in coils or in straight lengths depending upon pipe size and material selected.

3. Open trench pipelines must allow for the jointing, cooling and snaking of the pipe. When using 'normal' trench widths, this can mean greater inconvenience to traffic but allows flexibility to overcome unforeseen obstructions and also ensures the ability to bed and surround the pipe properly. Narrow trenching with PE has the considerable advantages of reduced reinstatement costs and less spoil to handle. Not all subsoils are conducive to such a technique and proper laying, bedding and compaction is not always possible at the required depths of cover.

4. The flexibility of PE allows the accurate alignment of the pipeline to awkwardly contoured kerb races on housing sites. The reinstatement or replacement of pipes in established areas will minimize disruption for major cost advantages.

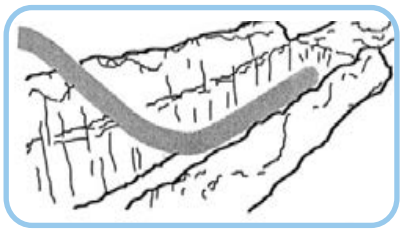
Installation Considerations

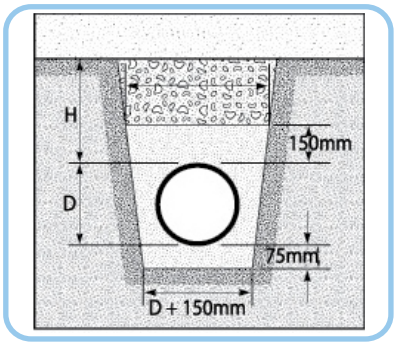
Expansion and Contraction

PE Pipe will expand or contract if it is installed during very hot or very cold weather, so it is recommended that the final pipe connections be made when the temperature of the pipe is stabilized at a temperature close to that of the backfilled trench.
PE lines can be laid directly on the natural surface by snaking the pipe during installation and allowing the pipe to move freely in service. Where the final joint connections are made in high ambient temperature, sufficient pipe length must be allowed to permit the pipe to cool, and hence contract, without pulling out of non end load bearing joints. HDPE pipes will expand or contract by 0.2mm/m/ oC rise or fall in temperature.

Trenching/Narrow/Chain

With the PE pipe string jointed above the ground the need to trench joint is eliminated and the width of the excavation can be minimized thus reducing labour, the amount of backfill required and reinstatement costs.
The maximum trench width should be as restricted as possible depending on the soil conditions. This is necessary for both economics and to develop side support. The width of trenches used with PE pipe may be reduced by jointing above ground in the case of butt or electro-fusion welding and then feeding the jointed pipe into the trench.





Trench Depths

The recommended minimum trench depth is determined by the loads imposed on the pipe such as the mass of backfill material, the anticipated traffic loads and any other superimposed loads. The depth of the trench should be sufficient to prevent damage to the pipe when the anticipated loads are imposed upon it.

Minimum Cover above the pipes

Trenches should be excavated to allow for the specified depth of bedding, the pipe diameter and the minimum recommend cover and overlay plus backfill above the pipes. Table below provides recommendations for minimum cover to pipe crown.

The cover requirements mentioned in the table here will provide adequate protection for all pressure ratings of pipe. Where it is necessary to use lower covers, several options are available.

Loading	Cover (mm)
Roads and streets	750
Driveways/similar areas, subject to traffic	600
Footpaths, gardens	500
Construction traffic	750

- Provide additional structural load bearing bridging over the trench. Temporary steel plates may be used in the case of construction loads.
- Use a high quality granular backfill e.g. crushed gravel or road base.
- Use a higher class of pipe than required for normal pressure or other considerations.

Bedding Material

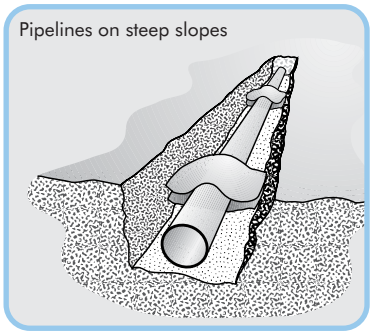
Preferred bedding materials are used as follows:

- Suitable sand, free from rock /other hard or sharp objects that would be retained on a 13.2mm sieve.
- Crushed rock or gravel evenly graded up to a maximum size of 20 mm.
- The excavated material may provide a suitable pipe underlay if it is free from rock or hard matter.

The suitability of a material depends on its compatibility. Granular materials (gravel or sand) containing little or no fines, or specification graded materials, requiring little or no compaction, are preferred.

Trenched Pipelines On Steep Slopes

When pipes are trenched on steep slopes, i.e. slopes steeper than 20% (1:5), trenched backfill around the pipe may be scoured out by water movement. Clay stops or sandbags should be placed at appropriate intervals above and below the pipe to stop erosion of the backfill.



Pipeline Buoyancy

Pipe under wet conditions can become buoyant in the trench. Pipes should be covered with sufficient overlay and backfill material to prevent inadvertent flotation and movement. A depth of cover over the pipe of 1.5 times the diameter is usually adequate.

Magnitude and the duration of the field pressure test (FPT)

All completed PE pipelines should be tested to ensure that all joints, fittings, anchorage blocks are installed correctly and that there are no defects in the pipes causing to leakage or loss of fluid.
PE pipelines may be tested as a single section for lengths up to 800 meters, depending on the size of the pipe and the availability of test water. Beyond this length, the pipeline should be progressively tested in sections.

Duration of pressure test for checking the integrity of pipe system is 1 hour. For PE pipelines, the recommended test pressure is as follows.

Testing pressure is 1.5 times the Working Pressure of the pipe system (BUT IT SHOULD NOT EXCEED THE MAXIMUM RATED PRESSURE OF THE PIPE). (For example, max. rated pressure of PN-8 pipe is 8 bar).

For the main test, the results are deemed to be satisfactory when the pressure loss observed from the PE pressure pipelines is $\leq$ 0.1 bar per hour over the test period.

On reaching the test pressure and satisfying the condition for minimal air entrapment, the pipeline is isolated, and the pressure allowed to decay.

Note that this pressure is usually slightly less than initial pressure due to the normal expansion of the pipeline under pressure, but no additional “top-up” pressure should be applied.

If at any stage during the pressure test an unacceptable leak is indicated, it is advisable to check all the mechanical fittings before visually inspecting the fusion joints. Any defect in the installation revealed by the test should be rectified and the test repeated. On completion of the test sequence, the remaining pressure should be released slowly until the pipeline returns to its pre-test condition.

Final Back Filling

After successful pressure test of pipeline, final back should be done. The material used for upper layers can be coarser than that used for the initial backfill. These layers should be carefully rammed down one on top of the other, until the backfill is completed.

Chemical Resistance Chart*

Common chemicals resisted by polyethylene pipes are listed below where

A = Very Good B = Good C = moderate D = Not recommended

Sr. No.	Chemicals	PE
01	Acetaldehyde	C
02	Acetamide	A
03	Acetic Acid 80%	D
04	Acetone	B
05	Acetylene	A
06	Alcohols: Amyl	B
07	Benzyl	D
08	Butyl	A
09	Ethyl	B
10	Isopropyl	A
11	Methyl	A
12	Aluminum Suphate	A
13	Ammonia	C
14	Aniline	B
15	Aromatic Hydrocarbons	C
16	Arsenic Acid	B
17	Barium Carbonate	B
18	Barium Sulphate	B
19	Copper Suphate	B
20	Diesel Fuel	C
21	Ethylene Glycol	A
22	Fatty Acids	A
23	Ferric Chloride	A
24	Ferric Sulphate	A
25	Flourine	C
26	Formaldehyde 100%	B
27	Formic Acid	B
28	Gasoline	C
29	Heptane	B
30	Hyderochloric Acid 20%	A
31	Hydrogen Peroxide	C
32	Iodine	A

Sr. No.	Chemicals	PE
33	Benzaldehyde	A
34	benzene	C
35	Benzonic Acid	B
36	Benzol	C
37	Borax	A
38	Boric Acid	A
39	Butadiene	D
40	Butane	C
41	Butylene	B
42	Calcium Sulphate	B
43	Butylene	B
44	Carbon Dioxide	C
45	Carbon Disulfide	C
46	Carbonic Acid	B
47	Chlorine, Anhydrous	B
48	Chloroform	C
49	Chromic Acid 50%	A
50	Citirc Acid	A
51	Magnesium Hydroxide	A
52	Mercury	A
53	Oleum 100%	D
54	Petrolatum	B
55	Phenol	B
56	Phosphoric Acid	B
57	Potassium Carbonate	A
58	Silver Nitrate	B
59	Sodium Bicarbonate	A
60	Stearic Acid	B
61	Sulphuric Acid	B
62	Tannic Acid	B
63	Toluene	C
64	Zink Sulphate	A

Dadex provides a complete range of fittings for its Polyethylene (PE) pipes, manufactured in different diameters and grades. These fittings are manufactured as per international standards to provide leak free and reliable jointing system. These fittings are either compression type or Butt fused type depending upon the diameter of pipes. Butt fused fittings are further sub-divided into Injection moulded and Fabricated type. Fittings are compatible with polyethylene pipes manufactured as per IS04427 & DIN 8074/75.

Fittings are continuously monitored throughout the entire production process in accordance with the internal testing programs in compliance with the standards EN 1555, EN 12201, ISO 4427.

The barcode label applied on the electrofusion fittings contain the welding parameter according to standard. ISO 13950 and the traceability data of the fitting according to standard ISO 12176-4.

It's very important to control thermal movement and the temperature and pressure relationship within a pressure system.

Due to the fine surface structure, injection moulded PE fittings are ideal for transporting gases and liquids. The excellent UV resistance offers the possibility for outside installation without further need for additional protection effort.

Material

Polyethylene (PE100) classified according to minimum long-term strength (MRS), Colour black.

Geometric Properties

Outside diameter and wall thickness according to EN 1555 and EN 12201.
Tolerances of z-sizes depend on the dimension.
Up to d/D 160 = +/-3 mm and
from d/D 180= +/- 10 mm.

Joining Technology

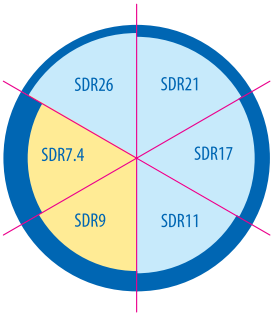
- Butt-Welding
- Electro fusion socket-welding
- Flange connection

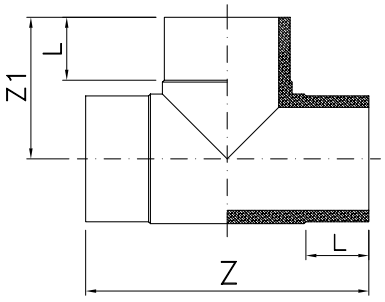


Equal Tee 90° (Injection Moulded)

Nominal size (dn) (mm)	Z (mm)	Z1 (mm)	L (mm)
*20	143	74	41
*25	158	78	41
32	166	82	45
40	188	96	50
50	210	107	50.5
63	219	115	66
75	246	129	71
90	271	143	70
110	318	167	85
125	335	168	90
*140	393	192	92
160	400	200	101
180	465	233	106
200	448	239	108
225	454	262	102
250	517	282	100
280	600	323	94
315	595	298	92
355	650	325	121
400	645	325	100
450	970	510	200
500	970	510	200

- *Producing up to size 315 mm with SDR 9 and SDR 7.4*
- * *Future Products*





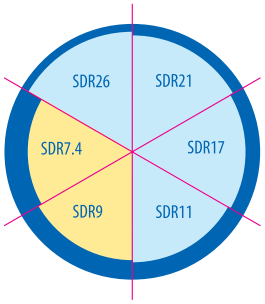


Reduced Tee 90° (Injection Moulded)

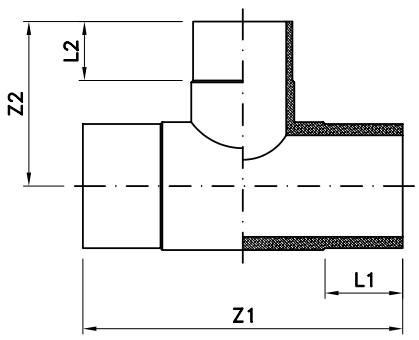
Nominal size (mm)	L1 (mm)	L2 (mm)	Z1 (mm)	Z2 (mm)
63×40	66	49	217	105
63×50	67	63	220	109
75×40	72	50	249	119
75×50	71	63	246	121
75×63	72	66	250	126
90×63	71	67	271	133
90×75	69	70	267	128
110×63	87	66	318	168
110×75	81	72	320	159
110×90	86	80	318	155
125×63	89	66	336	166
125×75	89	71	337	167
125×90	89	81	332	163
125×110	90	86	340	168
160×63	102	67	404	197
160×75	98	75	400	195
160×90	101	83	398	201
160×110	102	88	405	204
200×63	116	72	440	220
200×75	115	72	440	220
200×90	117	83	448	228
200×110	104	79	435	215

Nominal size (mm)	L1 (mm)	L2 (mm)	Z1 (mm)	Z2 (mm)
200×160	116	90	450	214
250×90	103	79	515	267
250×110	102	87	515	270
250×125	101	95	515	268
250×160	102	103	515	270
250×200	100	101	515	268
315×90	92	92	595	296
315×110	92	92	595	296
315×125	92	92	595	296
315×160	87	87	580	289
315×200	91	95	590	301
315×250	91	103	585	296
355×110	118	87	640	290
355×160	118	102	640	307
355×200	119	119	644	332
355×250	120	113	645	324
400×160	98	100	645	335
400×200	99	102	643	337
400×250	98	100	650	330

- Producing up to size 315mm with SDR9 and SDR 7.4
  - Other branches with make by Reducer.



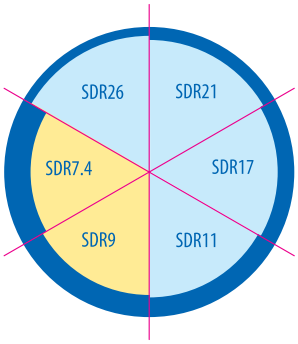


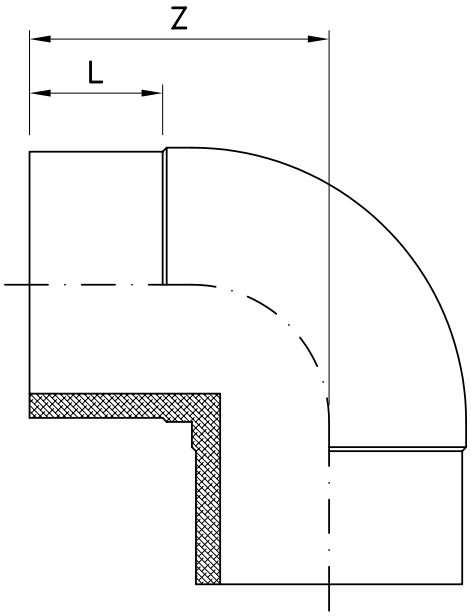


Elbow 90° (Injection Moulded)

Nominal size (mm)	Z (mm)	(mm)
*20	63	41
*25	63	41
32	72	45
40	80	50
50	110	60
63	112	61
75	120	60
90	127	55
110	153	84
125	172	85
160	200	102
180	245	126
200	248	117
225	243	111
250	250	95
315	329	120
355	325	118
400	376	120
450	380	120
500	390	120

- *Producing up to size 315 mm with SDR 9 and SDR 7.4*
* *Future Products*



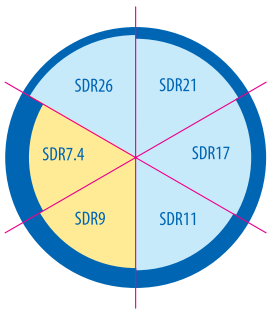




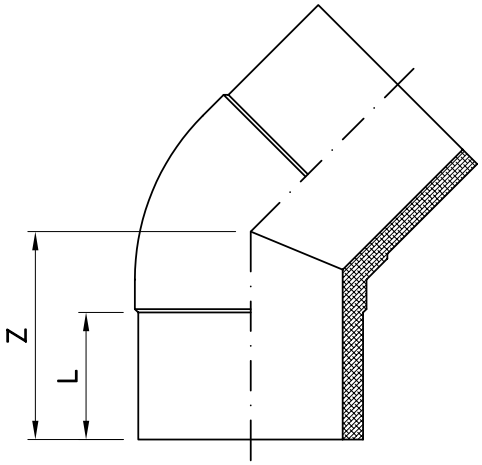
Elbow 45° (Injection Moulded)

Nominal size (mm)	Z (mm)	L (mm)
*25	63	41
*32	72	44
*40	80	49
50	92	58
63	94	58
75	105	60
90	120	79
110	121	85
125	130	88
160	148	85
180	166	106
200	180	117
225	184	111
250	210	128
315	206	102
*355	200	120
400	222	118

- *Producing up to size 315 mm with SDR 9 and SDR 7.4*
- * *Future Products*

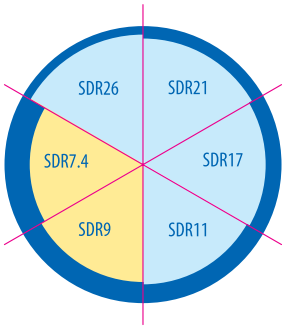


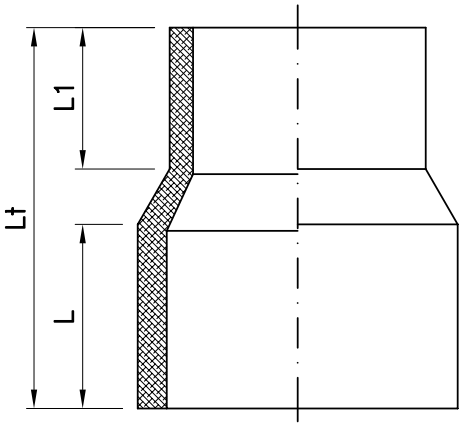




Concentric Reducer long spigot (Injection Moulded)

Nominal size (mm)	L1 (mm)	L (mm)	Lt (mm)
40×32	45	50	125
50×25	48	59	134
50×32	33	56	122
50×40	35	56	125
63×32	39	59	131
63×40	49	56	132
63×50	59	51	127
75×50	57	66	148
75×63	62	68	144
90×63	51	69	139
90×75	66	69	152
110×20	41	77	154
110×32	41	77	154
110×63	54	80	168
110×75	54	77	167
110×90	64	82	167
125×90	73	91	210
125×110	63	74	151
160×90	87	76	212
160×110	83	101	227
160×125	70	94	215
200×110	88	101	248
200×160	86	132	283
250×110	85	120	285
250×160	96	121	294
250×200	88	122	287
315×250	135	88	300
500×400			
500×450			
560×400			
560×450			
560×500	100	100	380
630×400			
630×450			
630×500			
630×560			







- *Producing up to size 315 mm with SDR 9 and SDR 7.4*

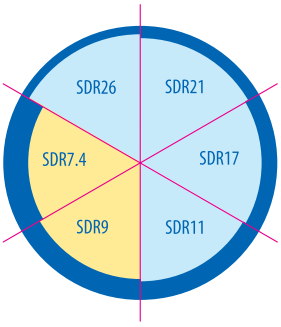
* *Future Products*

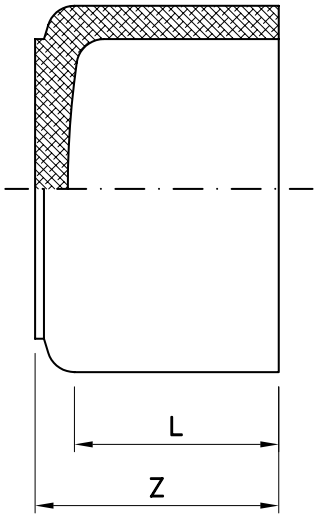
Cap (Injection Moulded)

Nominal size (mm)	Z (mm)	L (mm)
*25	50	40
*32	50	40
40	63	57
50	63	57
63	65	52
75	78	57
90	88	65
110	91	68
125	100	79
160	122	88
200	135	114
*250	205	120
*315	225	150

- *Producing up to size 315 mm with SDR 9 and SDR 7.4*

* *Future Products*





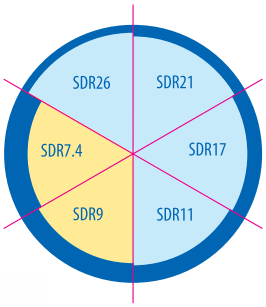


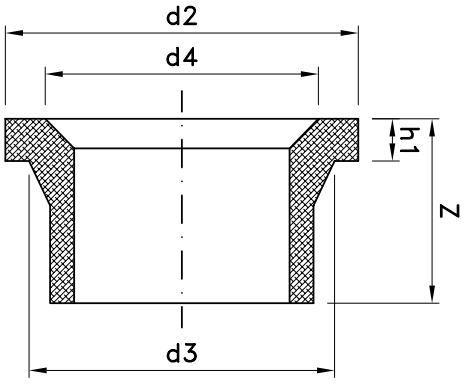
Flange Adaptor-Type B (Injection Moulded)

Nominal size (mm)	h1 (mm)		d2 (mm)	d3 (mm)	d4 (mm)	Z (mm)	
	PN10	PN16				Normal	Long
40	11	11	78	50	66	81	85
50	12	12	88	75	78	83	85
63	14	14	102	75	100	92	95
75	16	16	122	89	114	111	125
90	17	17	138	105	127	120	140
110	18	18	158	125	151	127	160
125	18	25	158	132	158	121	170
140	18	25	188	155	203	153	200
160	18	25	212	175	210	164	200
180	20	30	212	180	245	171	200
200	24	32	268	232	265	180	200
225	24	32	268	235	300	193	200
250	25	35	320	285	340	200	-
280	25	35	320	291	385	200	-
315	25	35	370	335	400	222	-
355	30	40	430	373	440	178	-
400	33	46	482	427	490	154	-
450	46	60	585	514	545	160	-
500	46	60	585	530	-	170	-
560	50	60	685	615	-	178	-
630	50	60	685	642	-	154	-
710	50	-	800	730	-	300	-
800	52	-	900	840	-	300	-
900	55	-	1005	944	-	300	-
1000	60	-	1110	1047	-	300	-

Could be weld with electrofusion coupler

Type B: *With chamfer suitable for butterfly valve .*



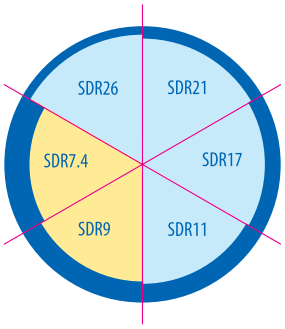




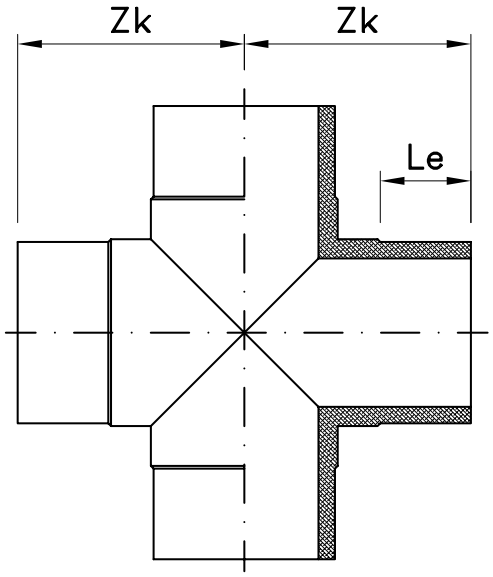
Cross 90° (Injection Moulded)

Nominal size (mm)	Zk (mm)	Le (mm)
90	149	80
110	165	85
*125	175	85
160	206	100
200	245	118
250	303	144
*315	330	155

- *Producing up to size 315 mm with SDR 9 and SDR 7.4*
- * *Future Products*



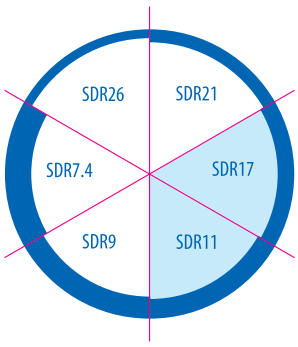


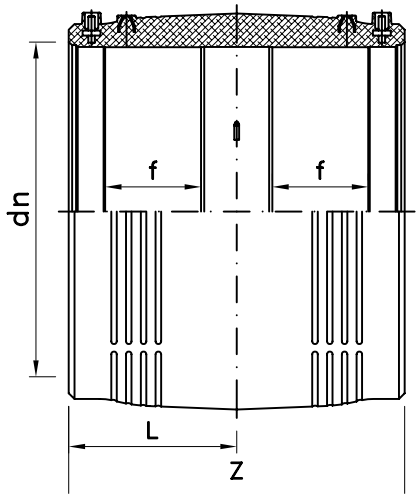


Coupler (Electrofusion Fittings)

Nominal size (dn) (mm)	L (mm)	f (mm)	Z (mm)
*20	41	10	82
*25	41	10	82
*32	44	10	88
*40	49	10	98
*50	55	10	110
63	47	17	95
75	52	21	106
90	61	26	123
110	71	36	144
125	78	39	157
160	88	51	177
200	100	59	204
225	110	72	221
250	115	60	233
315	130	67	262
*355	164	42	328
*400	179	47	358

SDR 26, SDR 21, SDR 9 and SDR 7.4 are produced according to customer order
** Future Products*





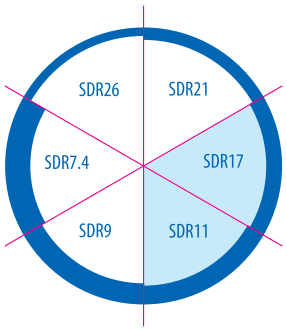


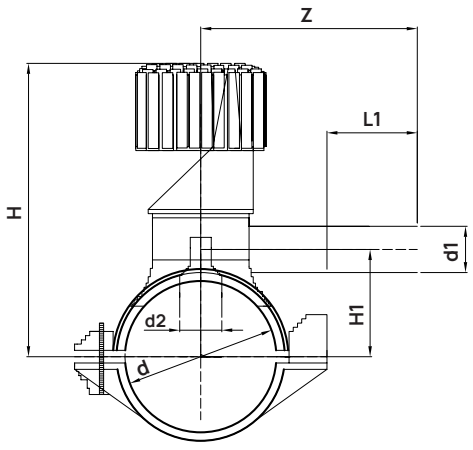
Tapping saddle with 360° Rotatable outlet (Electrofusion Fittings)

Nominal size (mm)	d2 (mm)	H (mm)	H1 (mm)	L (mm)	L1(mm)	Z(mm)
63×25	32	186	108	165	71	130
63×32	32	186	108	165	76	130
63×40	32	186	108	165	81	137
63×50	32	186	108	165	86	137
63×63	32	134	112	165	100	160
90×25	32	199	121	165	71	130
90×32	32	199	121	165	76	130
90×40	32	199	121	165	81	137
90×50	32	199	121	165	86	137
90×63	32	248	126	165	100	160
110×25	32	209	131	165	71	130
110×32	32	209	131	165	76	130
110×40	32	209	131	165	81	137
110×50	32	209	131	165	86	137
110×63	35	258	136	165	100	160
160×25	32	243	156	165	71	130
160×32	32	243	156	165	76	130
160×40	32	243	156	165	81	137
160×50	32	243	156	165	86	137
160×63	35	283	161	165	100	160
200×25	32	254	176	165	71	130
200×32	32	254	176	165	76	130
200×40	32	254	176	165	81	137
200×50	32	254	176	165	86	137
200×63	35	303	181	165	100	160
*250×25	32	279	201	165	76	130
*250×32	32	279	201	165	76	130
*250×40	32	279	201	165	81	137
*250×50	32	279	201	165	86	137
*250×63	35	328	206	165	100	160

* *Future Products*



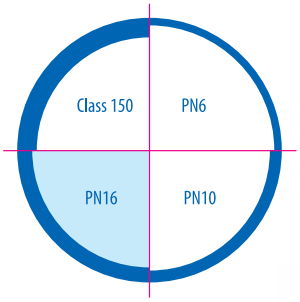


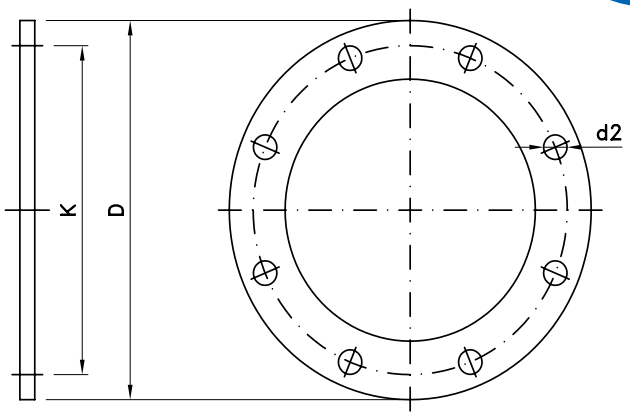


Backing Ring (EN 1092 - 2013) - PN16

DN	Pipe Outside (mm)	D (mm)	K (mm)	d2 (mm)	Bolting	
					Number	Size
20	**25**	105	75	14	4	M12
25	32	115	85	14	4	M12
32	40	140	100	18	4	M16
40	50	150	110	18	4	M16
50	63	165	125	18	4	M16
65	75	185	145	18	8	M16
80	90	200	160	18	8	M16
100	110	220	180	18	8	M16
100	125	220	180	18	8	M16
125	140	250	210	18	8	M16
150	160	285	240	22	8	M20
150	180	285	240	22	8	M20
200	200	340	295	22	12	M20
200	225	340	295	22	12	M20
250	250	405	355	26	12	M24
250	280	405	355	26	12	M24
300	315	460	410	26	12	M24
350	355	520	470	26	16	M24
400	400	580	525	30	16	M27
450	450	640	585	30	20	M27
500	500	715	650	33	20	M30
600	630	840	770	36	20	M33

Note :Thickness According to Customer Order





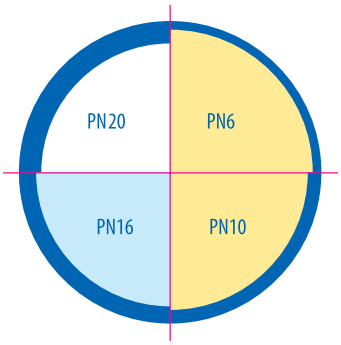


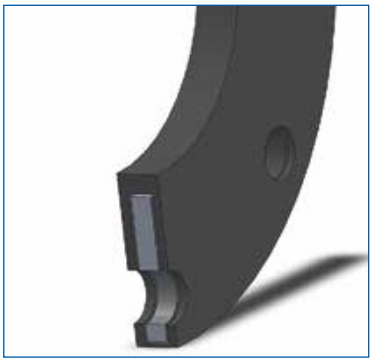
Backing Ring (PP/Steel Backing Ring)

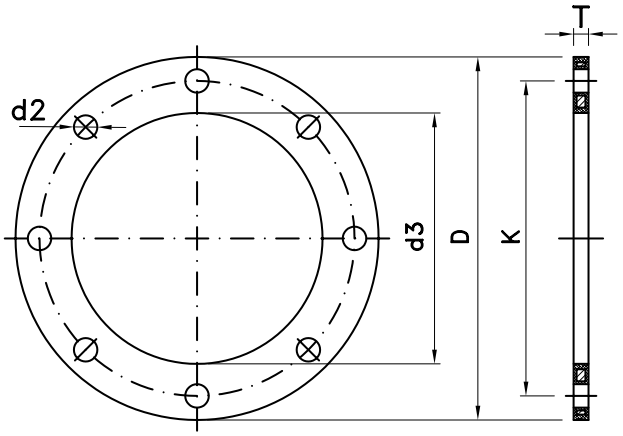
Flange Size (mm)	Code	D (mm)	d3 (mm)	K (mm)	d2 (mm)	Bolt Number	Bolt Size	T±1.5 (mm)
63	RI1C06	165	78	125	18	4	M 16	18
75	RI1C07	185	92	145	18	4	M 16	18
90	RI1C08	201	108	160	18	8	M 16	20
110	RI1C09	221	130	180	18	8	M 16	20
125	RI1C10	251	135	180	18	8	M 16	20
160	RI1C12	286	177	240	22	8	M 20	24
200	RI1C14	340	235	295	22	8	M 20	24
225	RI1C15	340	238	295	22	8	M 20	24
250	RI1C16	409	288	350	22	12	M 20	30
315	RI1C18	463	338	400	22	12	M 20	34
*355	RI1C19	520	376	470	26	16	M20	40
*400	RI1C20	580	430	525	30	16	M24	40

- *Possible To use For PN6 & PN10 .*
  Ring Steel : DIN2501

* *Future Products*



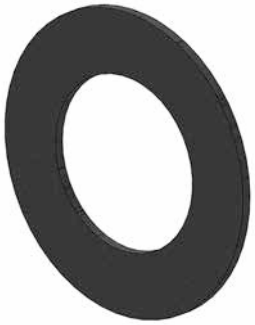


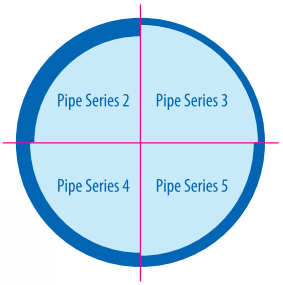


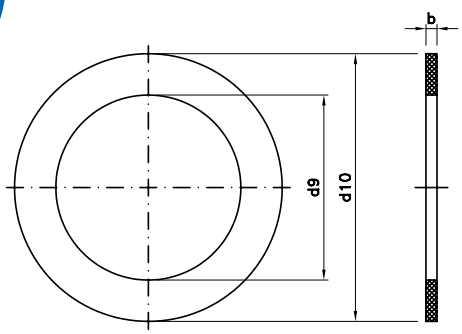


Gasket

Pipe Outside Diameter (mm)	d10 Outside of Gasket (mm)	Pipe Series (Inside Of Gasket) d9 (mm)				b (mm)
		2	3	4	5	
20	45	-	-	-	16	2
25	58	-	-	21	20	2
32	68	-	-	28	26	2
40	78	-	36	35	33	2
50	88	-	46	44	41	2
63	102	59	58	56	51	2
75	122	70	69	66	61	2
90	138	84	83	80	74	2
110	158	103	101	97	90	3
125	158	117	115	111	102	3
140	188	131	129	124	114	3
160	212	150	148	142	131	3
180	212	169	166	160	147	3
200	268	188	185	177	164	3
225	268	211	208	199	184	3
250	320	234	231	222	204	3
280	320	263	258	248	229	3
315	370	295	291	279	258	3
355	430	333	328	315	290	3
400	482	375	369	355	327	3
450	585	422	415	399	368	3
500	585	469	461	443	409	3
560	685	525	517	497	456	3
630	685	590	581	559	513	3
710	800	666	655	630	-	3
800	905	750	738	709	-	3
900	1005	844	831	798	-	3
1000	1110	938	923	885	-	3
1200	1330	1126	1108	-	-	3



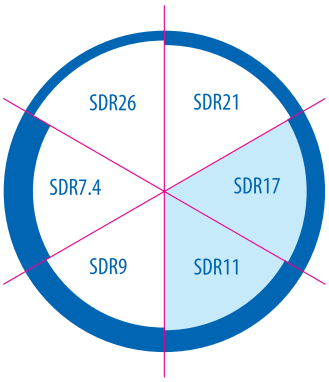


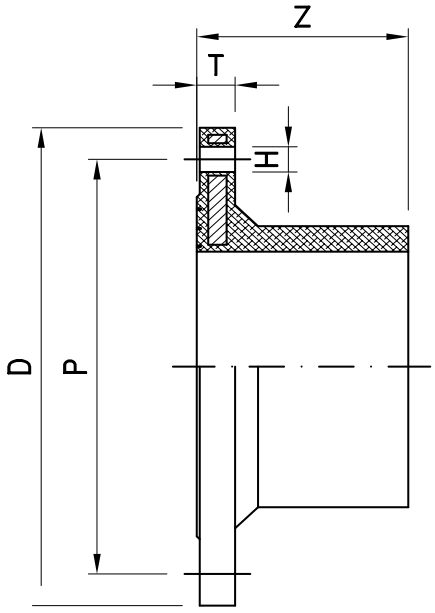


Steel Core Hole Flange (Flanged Fittings)

Nominal size (mm)	D (mm)		P (mm)	Z (mm)	T (mm)	H (mm)		Bolt Number
	PN16	PN10				PN16	PN10	
63	165	165	125	82	18	18	18	4
75	185	185	145	100	18	18	18	8
90	200	200	160	106	20	18	18	8
110	220	220	180	112	20	18	18	8
125	220	220	180	124	22	18	18	8
160	285	285	240	132	25	22	22	8
200	340	340	295	140	25	22	22	12-8
250	405	395	350	160	30	26	22	12
315	460	445	400	180	35	26	22	12
355	520	505	460	225	40	26	22	16
400	580	565	515	240	40	30	26	16

Could be weld with electrofusion coupler.
Ring Steel : DIN2501
**Future Products*





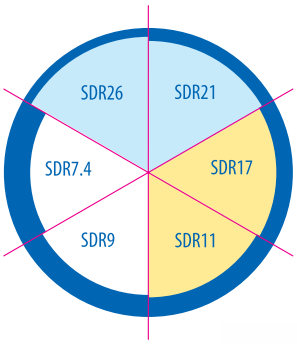


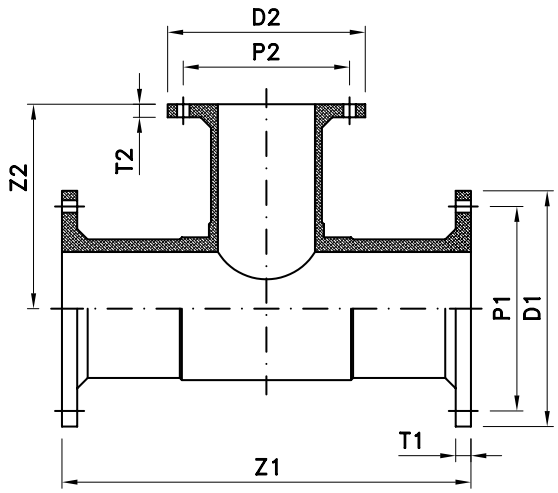
Flat Face Flanged Reduced Tee 90° (Flanged Fittings)

Nominal size (mm)	D1 (mm)	D2 (mm)	P1 (mm)	P2 (mm)	Z1 (mm)	Z2 (mm)	T1 ±2 (mm)	T2 ±2 (mm)	Bolt Number
110×75	217	183	180	145	360	174	20	18	8-4
110×90	218	195	180	160	360	172	22	22	8-8
160×75	278	183	240	145	495	248	22	18	8-4
160×90	278	195	240	160	495	248	22	22	8-8
160×110	278	218	240	180	495	248	22	20	8-8
200×90	335	195	295	160	593	303	22	20	8-8
200×110	340	219	295	180	590	301	22	20	8-8
200×160	340	278	295	240	590	301	22	22	8-8
250×90	395	200	350	160	600	300	26	20	12-8
250×110	395	218	350	180	600	300	26	20	12-8
250×160	395	282	350	240	585	295	28	22	12-8
315×90	444	200	400	160	595	300	30	22	12-8
315×110	444	218	400	180	595	300	30	22	12-8
315×160	444	280	400	240	600	302	30	24	12-8
315×200	444	338	400	295	595	300	30	26	12-8

PN4 bar and 6 bar : without backing ring

- *PN10 bar : With backing ring*
- *PN16 & 10 bar : Using steel core hole flange*





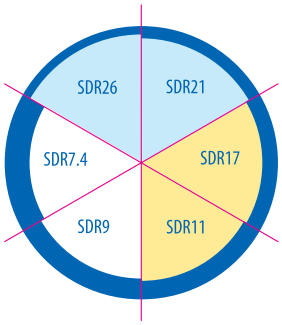


Flat Face Flanged Equal Tee 90° (Flanged Fittings)

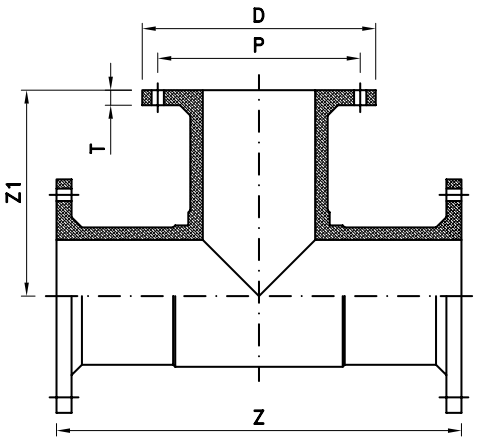
Nominal size (mm)	D (mm)	P (mm)	Z1 (mm)	Z (mm)	T ±2 (mm)	Bolt Number
90	200	160	148	309	20	8
110	215	180	165	360	20	8
125	245	180	214	430	20	8
160	280	240	247	500	20	8
200	342	295	300	595	22	8
250	395	350	295	600	26	12
315	445	400	171	595	30	12

PN4 bar and 6 bar : without backing ring

- *PN10 bar : With backing ring*
- *PN16 & 10 bar : Using steel core hole flange*





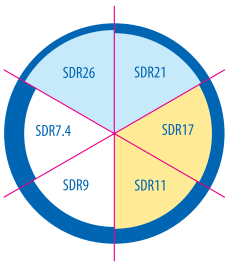


Flat Face Flanged Elbow 90° (Flanged Fittings)

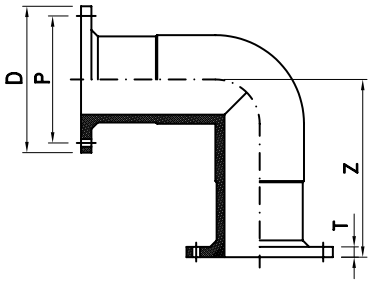
Nominal size (mm)	D (mm)	P (mm)	Z (mm)	T±2 (mm)	Bolt Number
90	200	160	152	18	8
110	215	180	183	20	8
125	243	180	218	20	8
160	280	240	300	22	8
200	340	295	280	22	8
250	395	350	300	24	12
315	445	400	300	28	12

PN4 bar and 6 bar : without backing ring

- *PN10 bar : With backing ring*
- *PN16 & 10 bar : Using steel core hole flange*







Collector (Special Fittings)





Special fittings like collectors, concentric reducer, etc. can also be produced according to customer order, technical and schematics.

DADEX Eternit Limited

Karachi Sales Office: Plot #.119-C, Office #.302, 3rd Floor, Anum Tower, Jami Commercial Street #: 13, DHA Phase-07, Karachi.
UAN: (92-21) 111 000 789, 021-36275087
Web : www.dadex.com www.facebook.com/dadexpk/

Dadex Sales Offices

- **Karachi:** (92-21) 36275087
  E-mail: sales@dadex.com.pk
- **Lahore:** (92-42) 3718 0152
  E-mail: lahore@dadex.com.pk
- **Faisalabad:** (92-41) 8861981
  E-mail: faisalabad@dadex.com.pk
- **Multan:** (92-61) 4545259
  E-mail: multan@dadex.com.pk
- **Islamabad:** (92-51) 5975109 - 110
  E-mail: islamabad@dadex.com.pk

For Technical information and assistance, please contact Dadex Techno Commercial Department at sales@dadex.com.pk

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Seller's Responsibility: The responsibility of the seller ceases once the goods are delivered to the buyer's representative at our factory, where delivery is taken in person. In other cases, the responsibility of the seller shall cease once the goods are delivered to the buyer's or carrier's authorized representative(s). No claims of any type—including in-transit loss, damage, pilferage, or short-delivery—will be entertained by the seller, and the buyer agrees to hold the seller harmless in this regard. Additionally, the seller shall not be responsible for any consequential damages, including but not limited to economic loss of any kind whatsoever, upon the products being delivered to the buyer as per the terms of this Clause. Any claim or responsibility as stated herein will not be entertained by the seller, and such action shall not be the cause of a dispute by the buyer.

Jan-2026